PROGRAMA DE PÓS-GRADUAÇÃO EM

Biologia Vegetal

Universidade Federal de Pernambuco

Proteínas reguladoras da arquitetura

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Leibniz Institute of Plant Genetics and Crop Plant Research

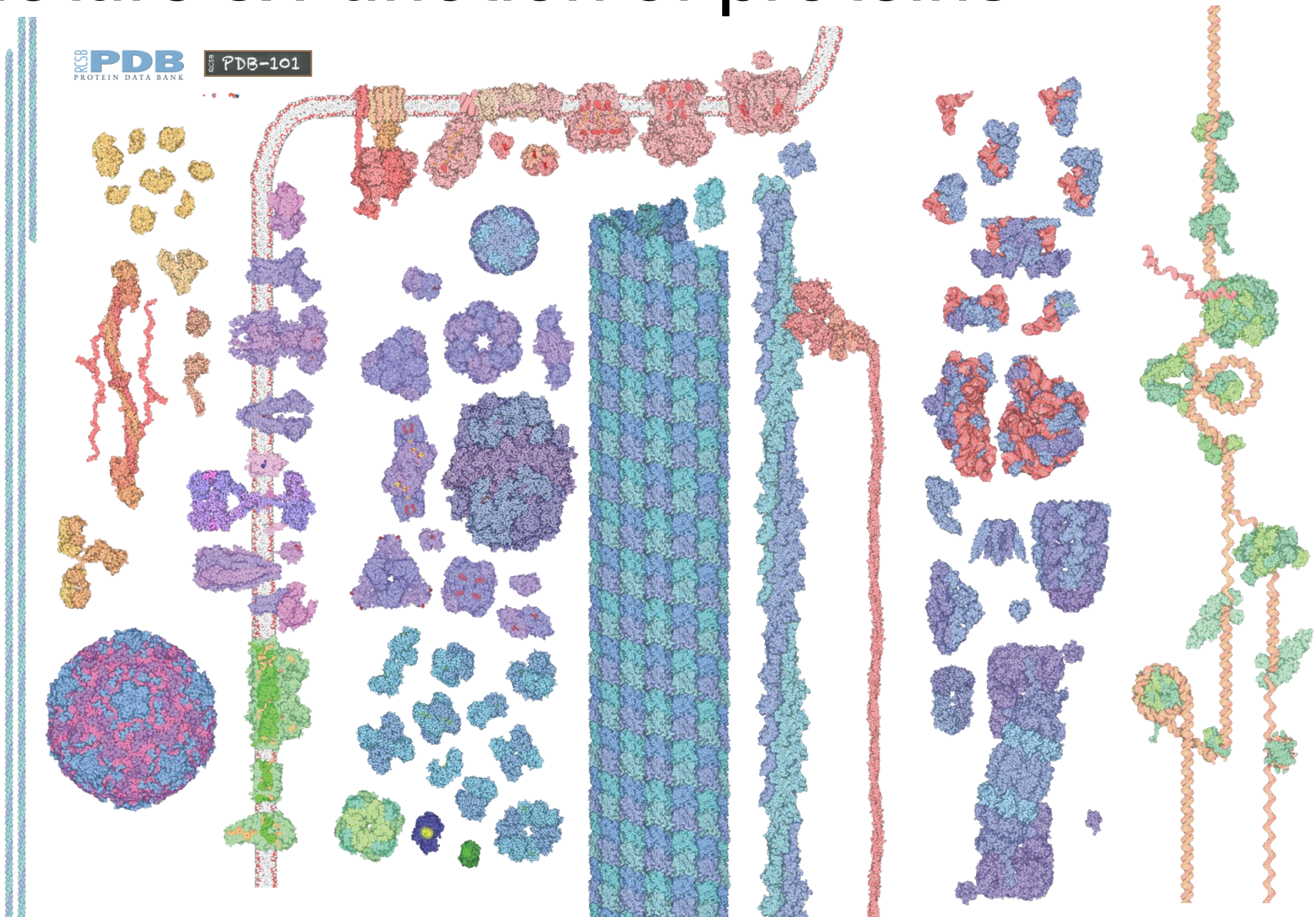


28 de outubro de 2026

In Silico Genebank Proteomics

IPK LEIBNIZ-INSTITUT

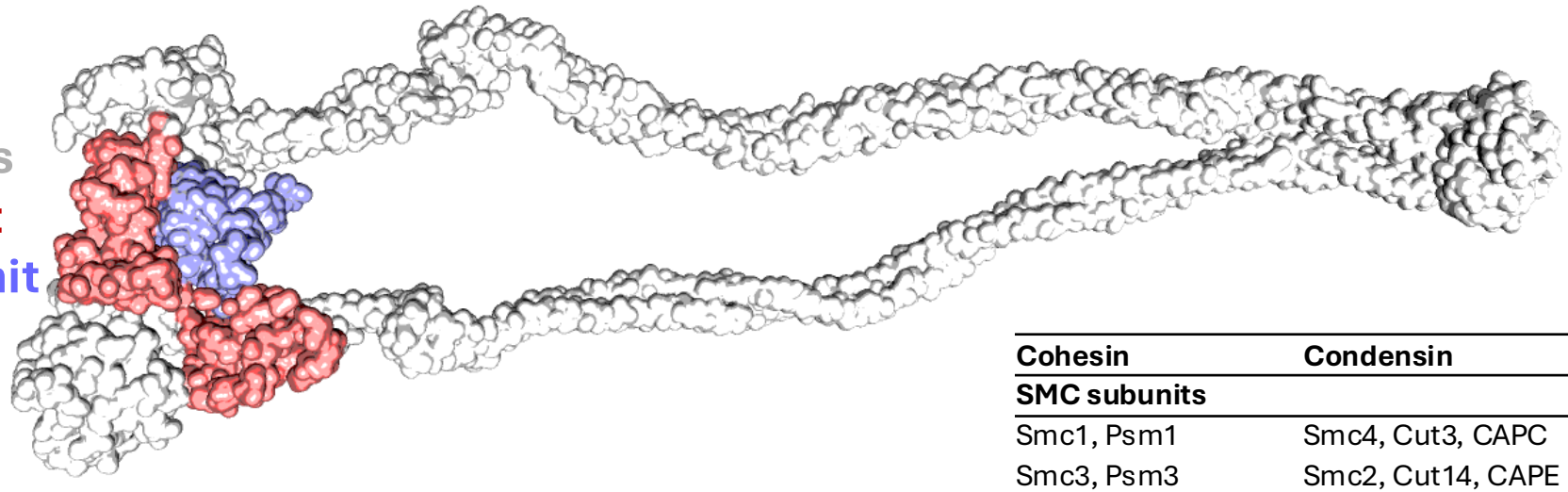
Structure & Function of proteins



Structural Maintenance of Chromosomes (SMC) complex proteins

SMC proteins act like chromatin loop extruders.

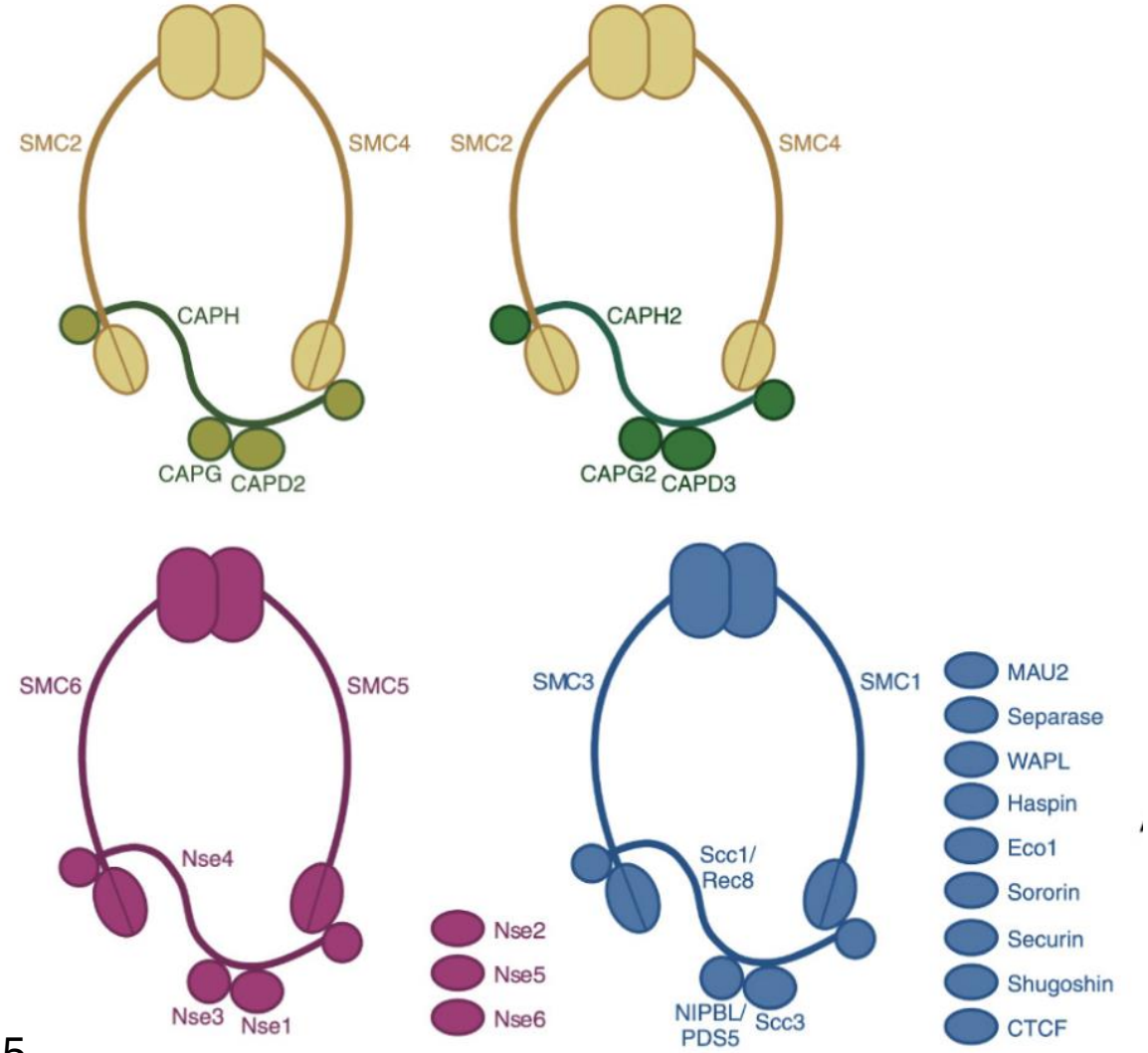
SMC subunits
HEAT subunit
Kleisin subunit



No pymol:
ma-ac7jp.pdb
ma-abc6s.pdb
Modelos feitos por Krepel et al. 2018

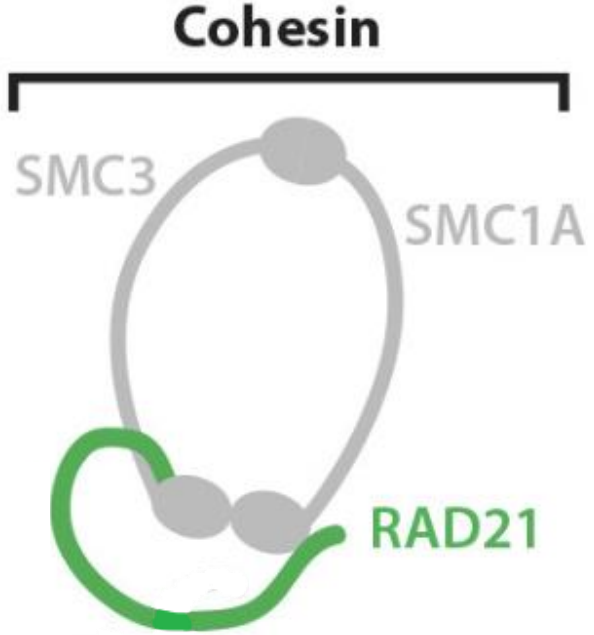
Cohesin	Condensin	Smc5-6 complex
SMC subunits		
Smc1, Psm1	Smc4, Cut3, CAPC	Smc5
Smc3, Psm3	Smc2, Cut14, CAPE	Smc6
Kleisin		
Scc1, Rad21, Rec8	Brn1,Cnd2, CAPH, CAPH2	Nse4, NSMCE4
HEAT subunits		
Scc3, Psc3, SA1, SA2	Ycs4, Cnd1, CAPD2, CAPD3	Nse5, NSMCE5
Pds5, PDS5A, PDS5B	Ycg1, Cnd3, CAPG, CAPG2	Nse6, NSCME6

Complexos diferentes

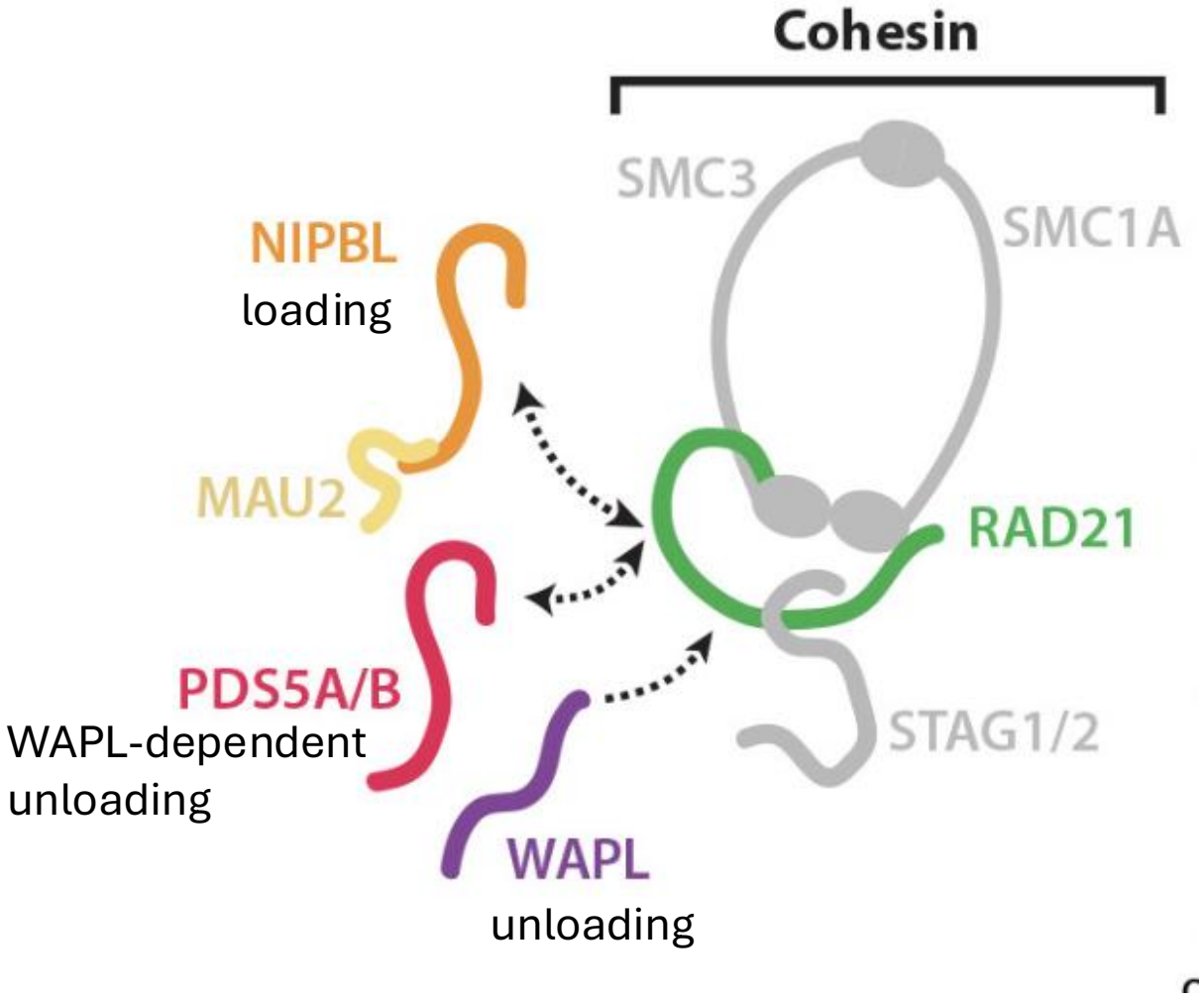


Van Hoff et al. 2025

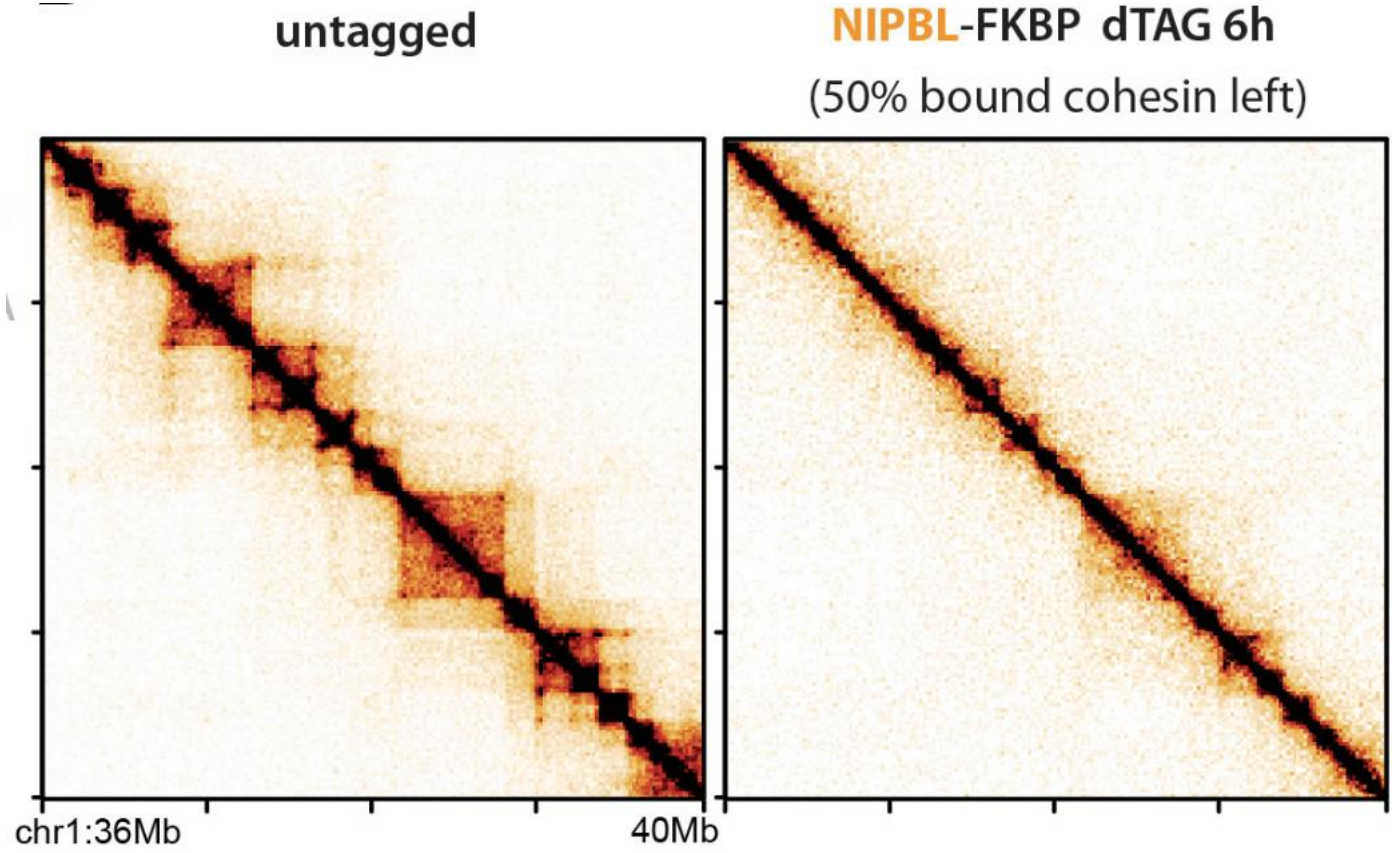
Role of each subunit of the complex



Role of each subunit of the complex



Depletion of NIPBL



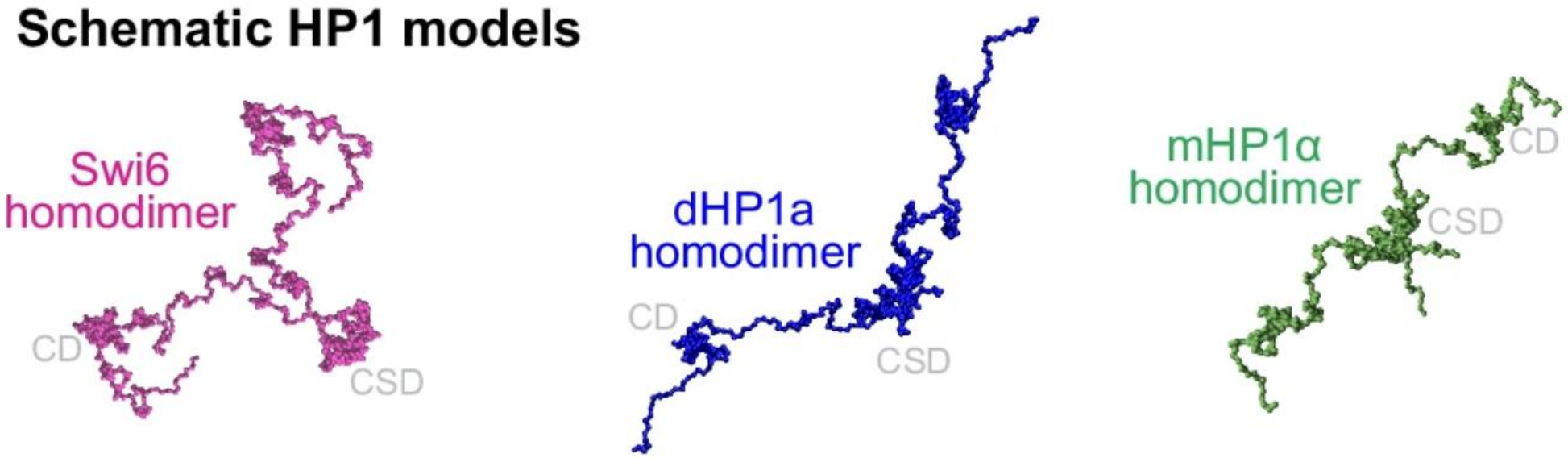
Shah et al. 2026

Coesina em ação

- https://youtu.be/-Cn7mOHAdFY?si=1MYBO-e7lr3cTLsw

Heterochromatin Protein 1

Proteína leitora de H3K9me



levedura mosca rato

Monômero de humano no AlphaFold: AF-P45973-F1

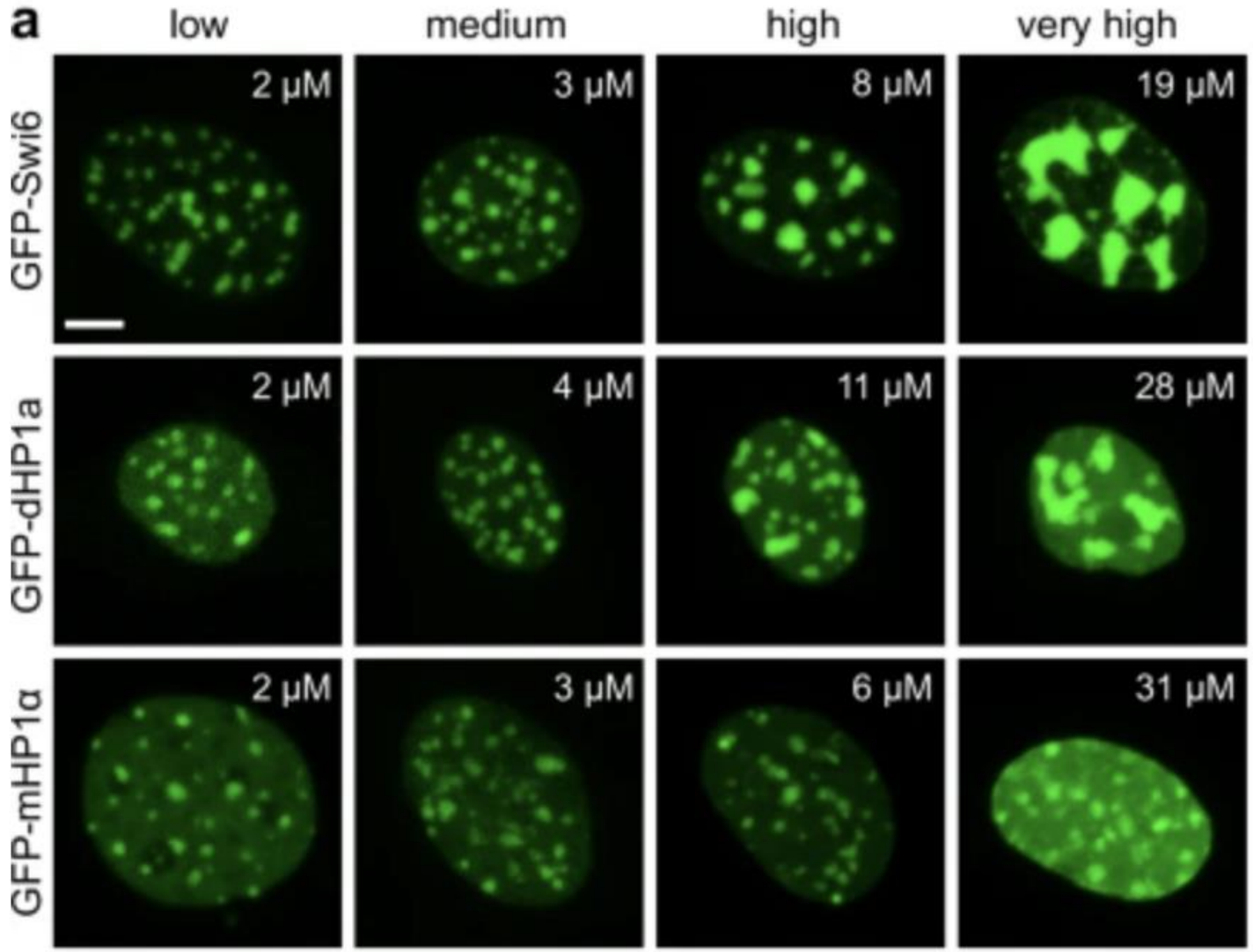
Bensaha et al. 2025

Separação de fase por HP1

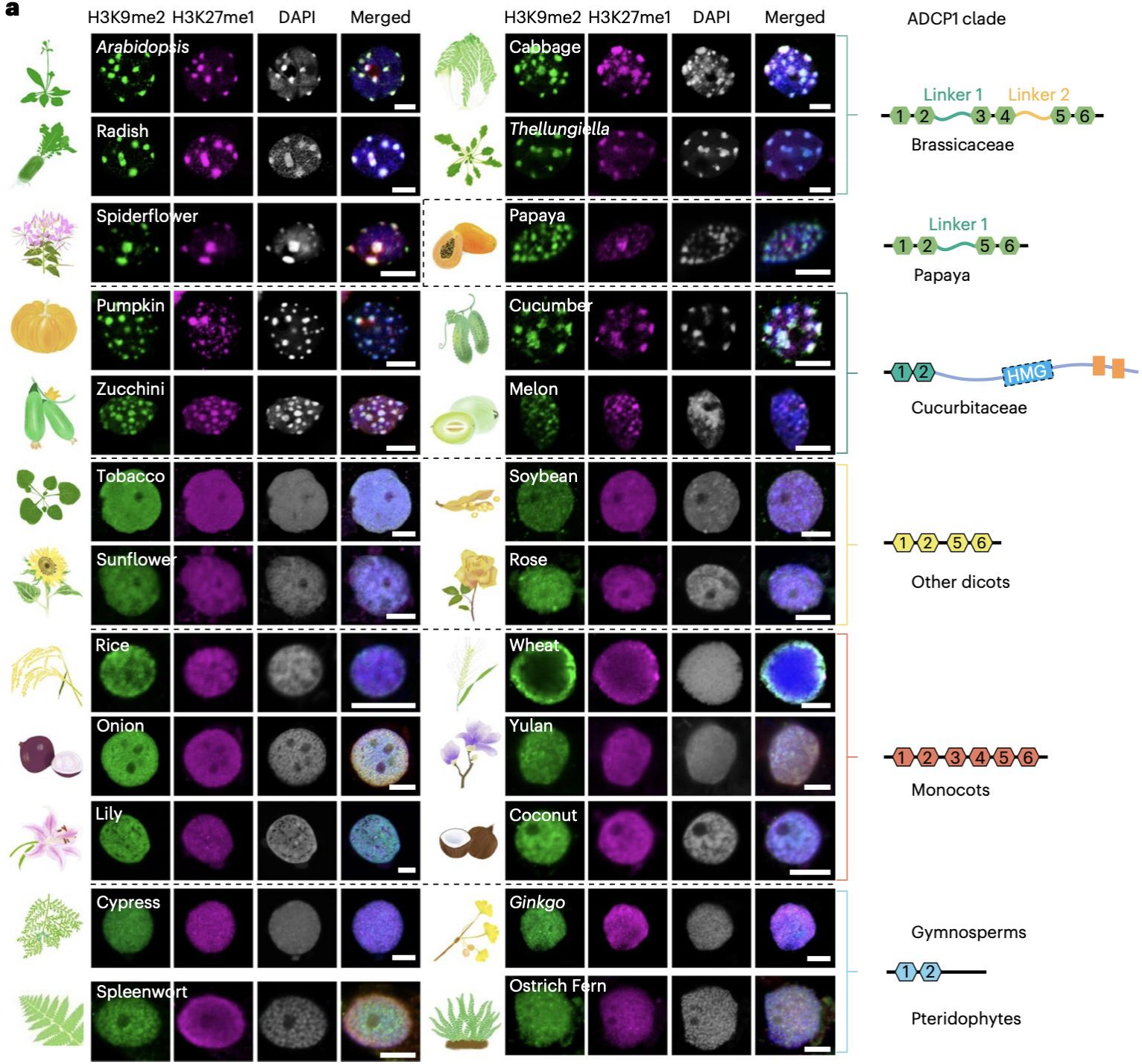
levedura

mosca

rato

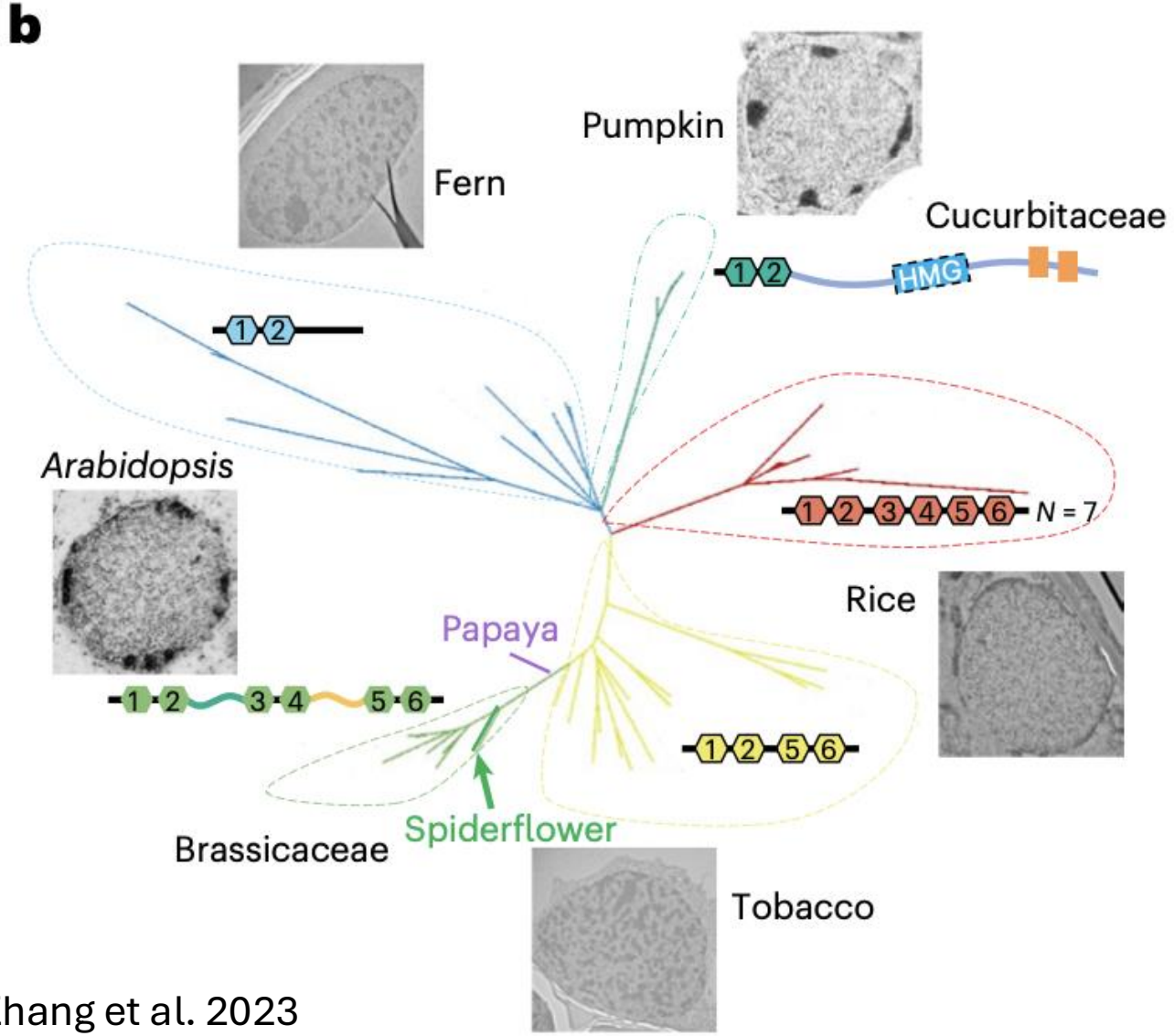


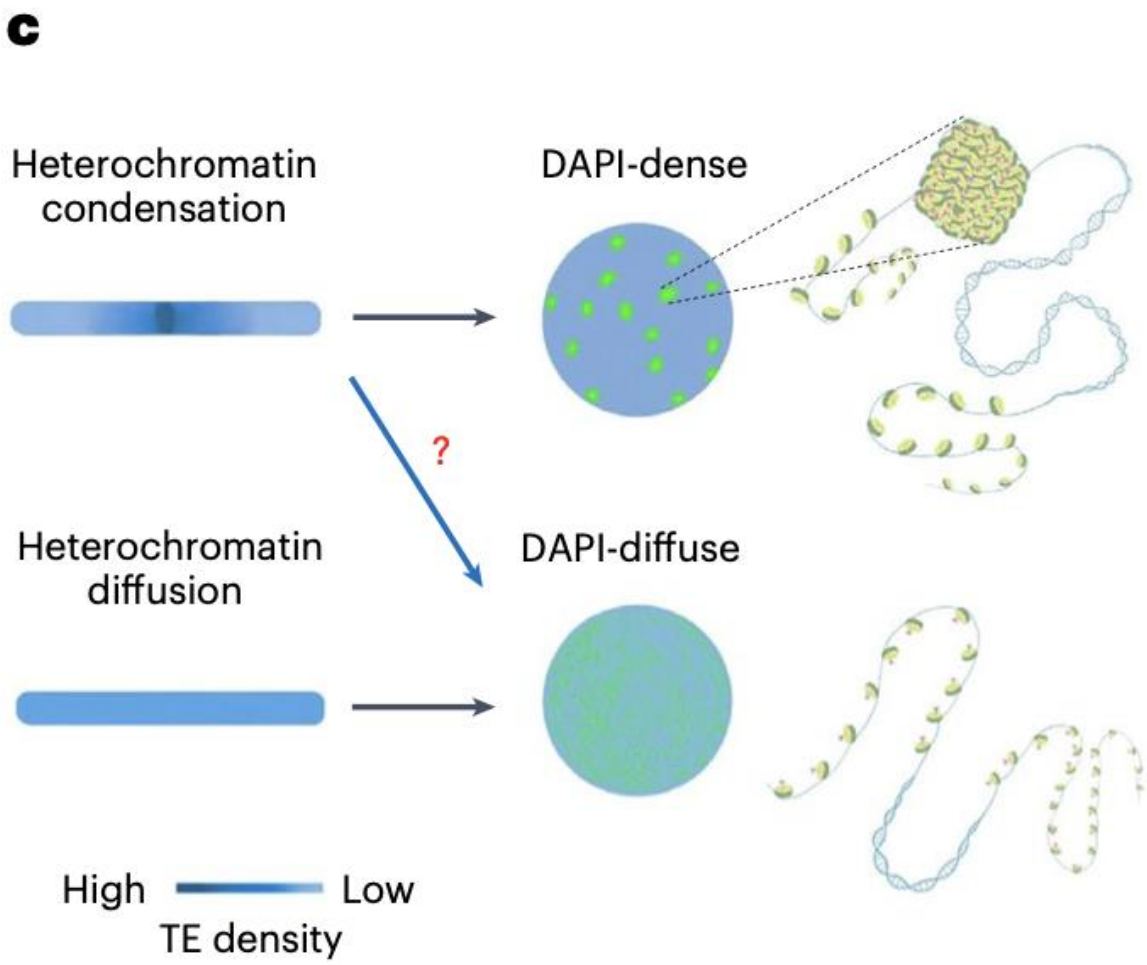
ADCP1 em plantas (Agenet domain-containing protein)



Evolução das ADCP1

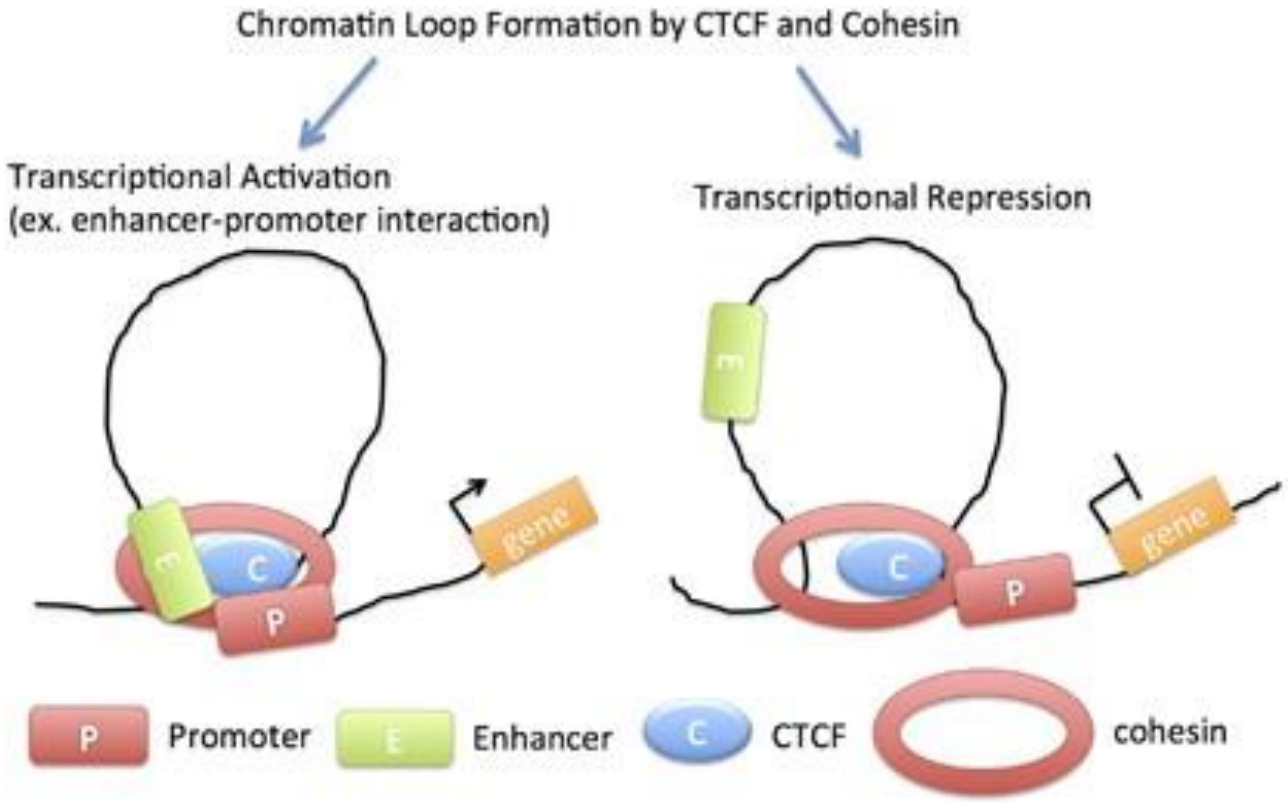
ADCP1 de Arabidopsis com cauda H3K9me1: 6ie4



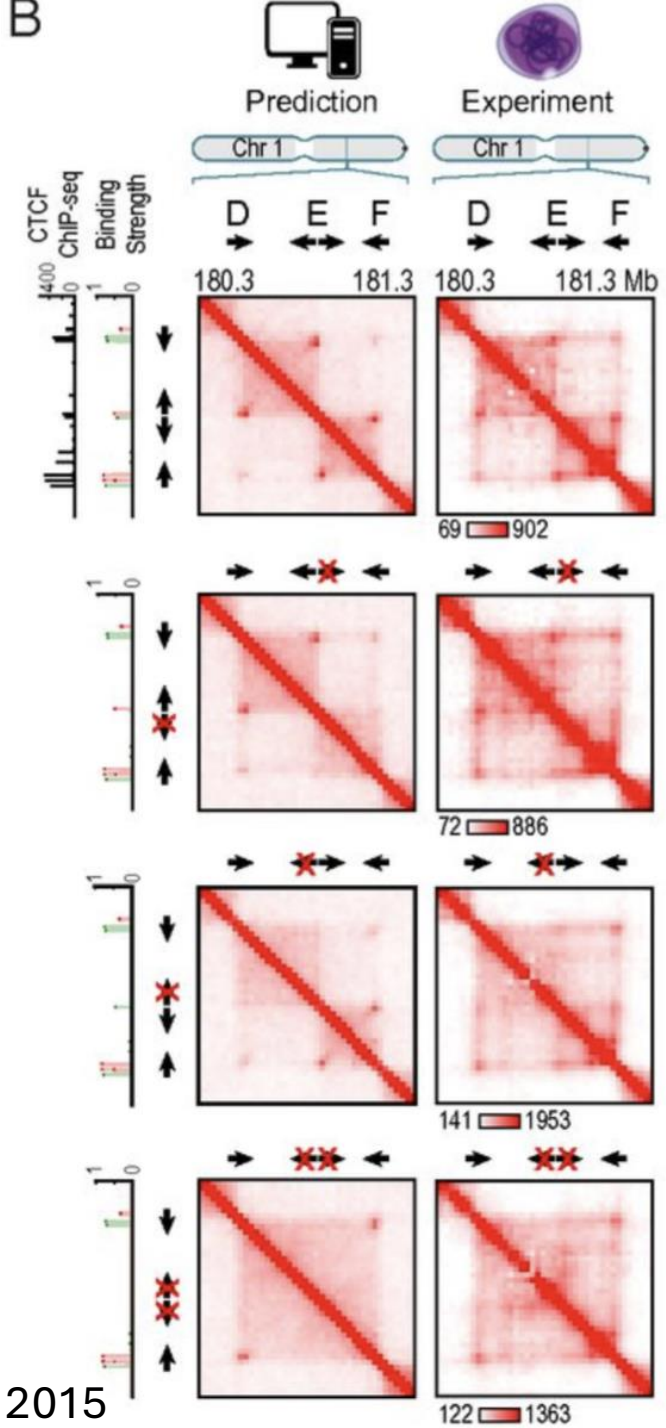


Zhang et al. 2023

CCCTC-binding factor (CTCF)

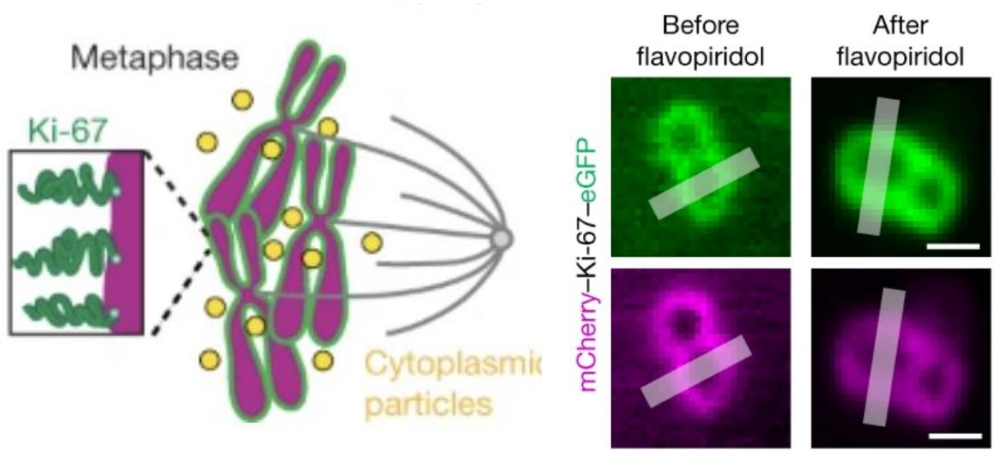


Kim et al. 2015



Sanborn et al. 2015

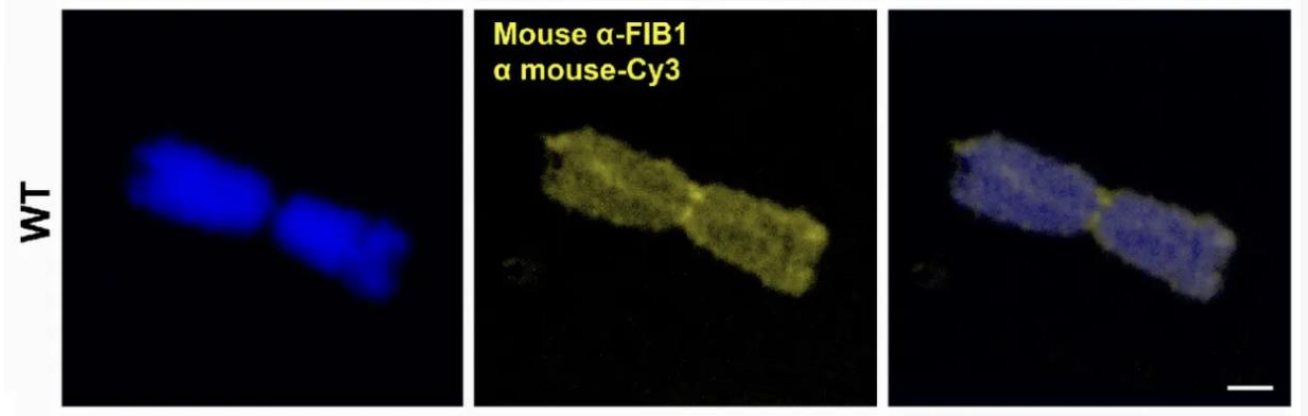
Casaco de proteínas em metáfase



Cuylen-Haering et al. 2020

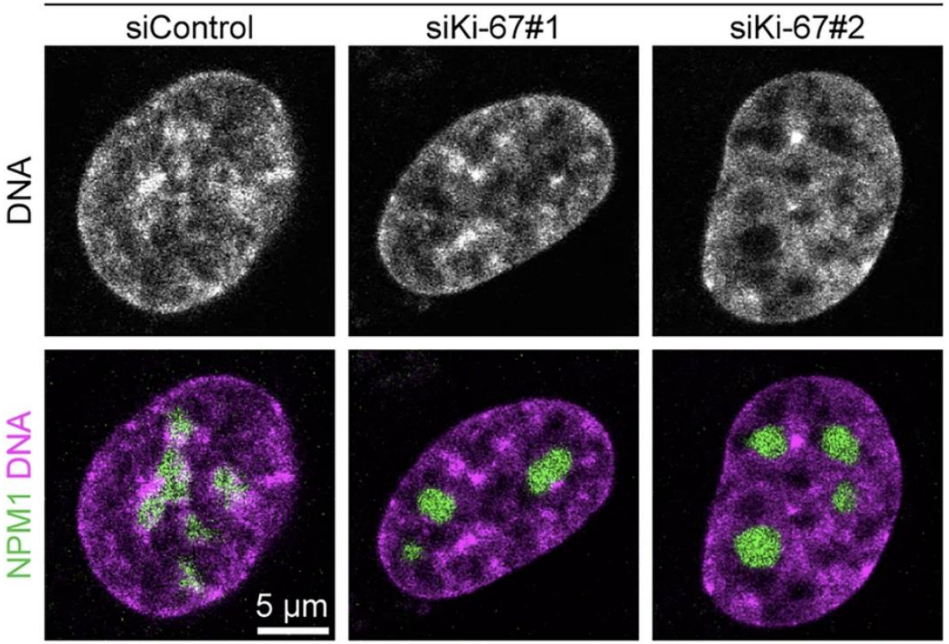
Plants não têm Ki-67

Camada pericromossomal



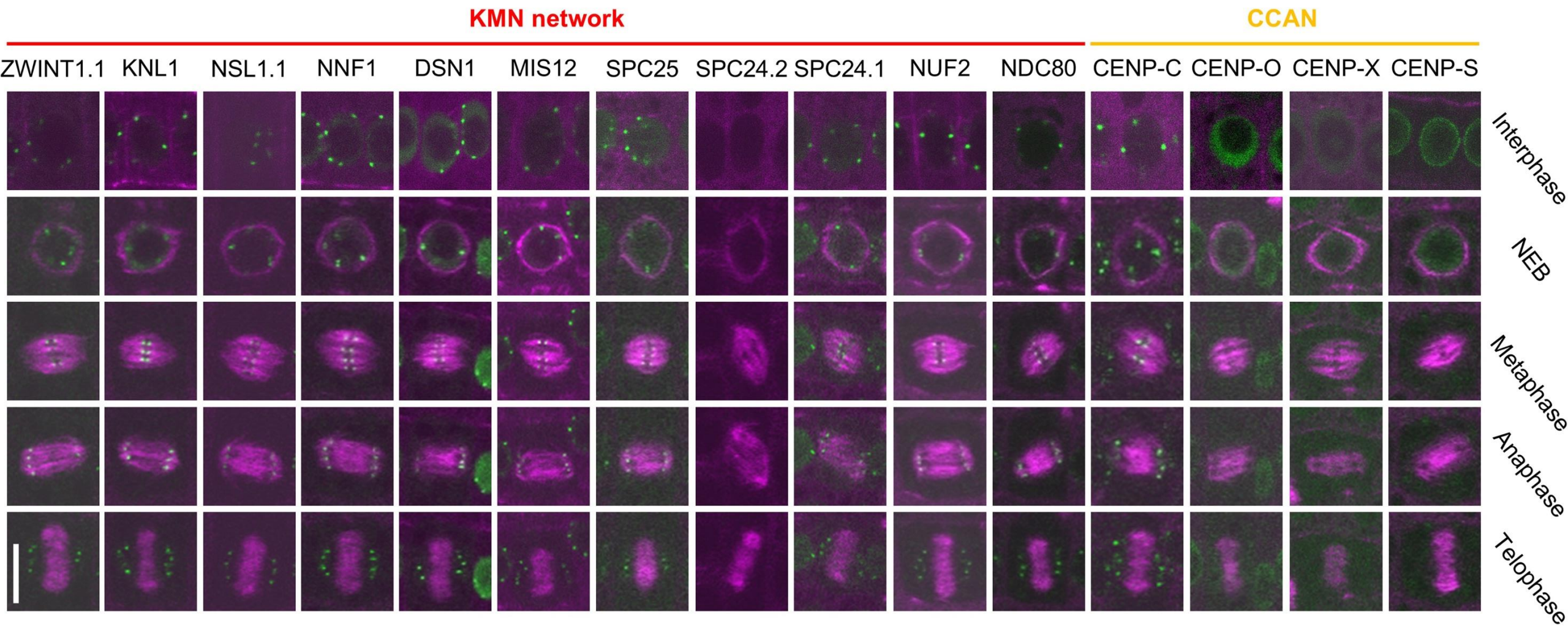
Cevada

Perutka et al. 2021



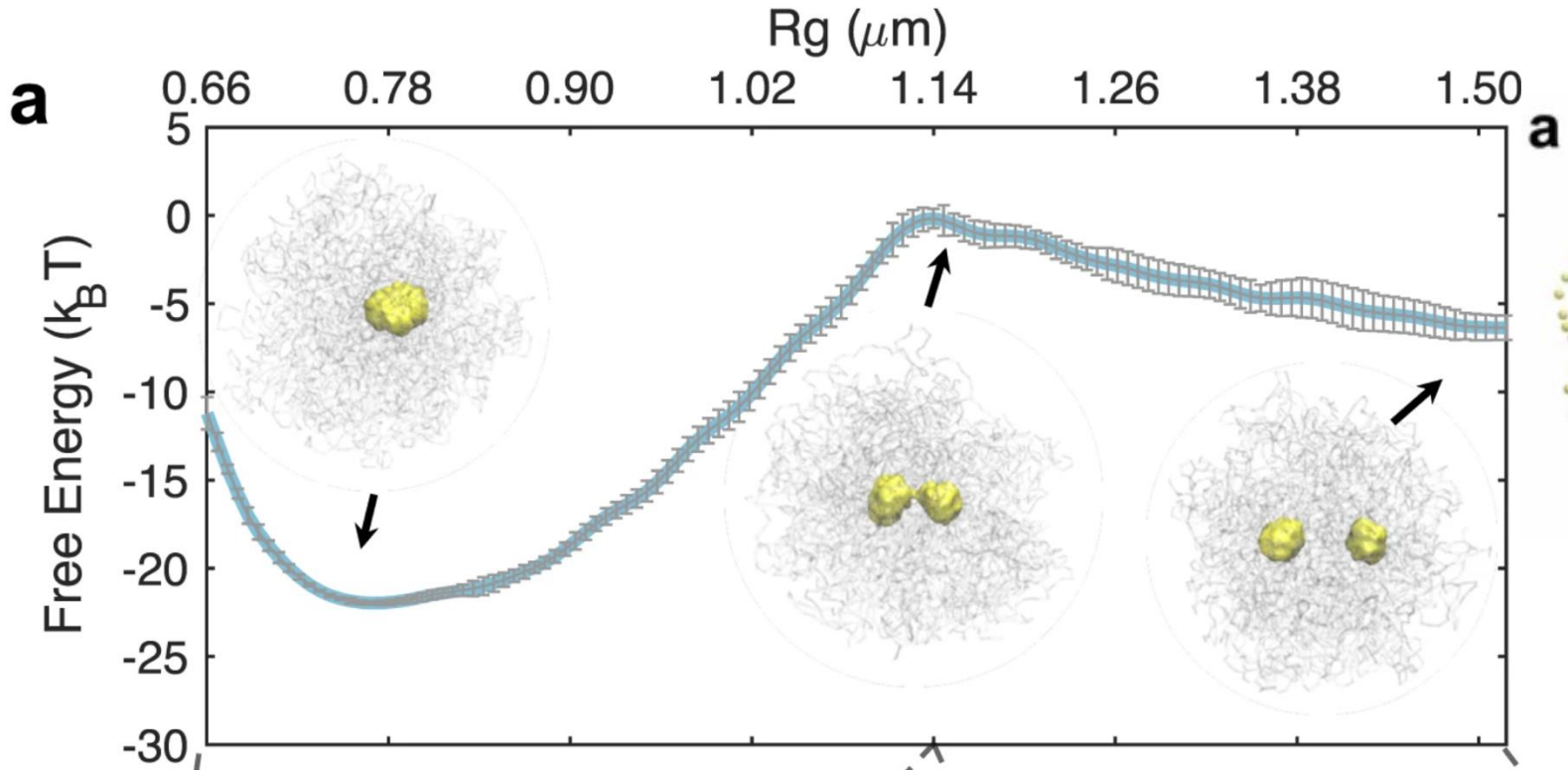
Schichler et al. 2026

Cinetócoro

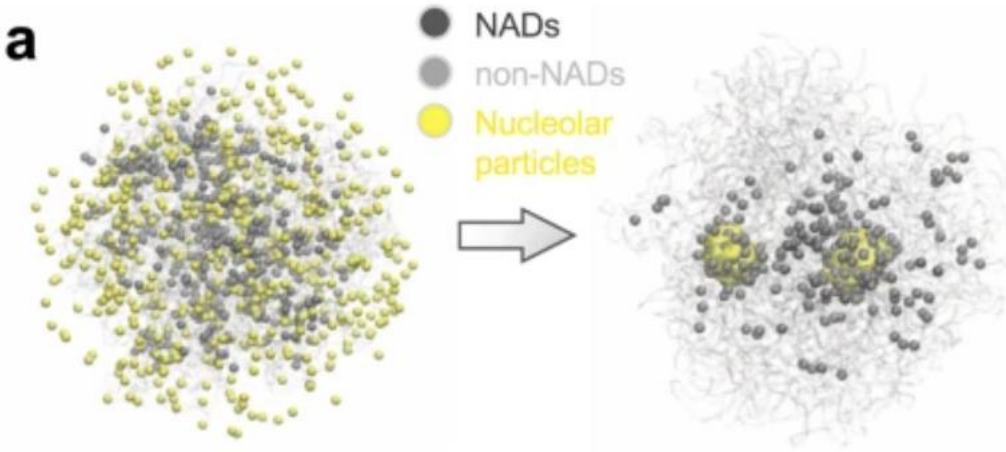


Pettkó-Szandtner et al . 2025

Nucleolar proteins



Coalescência de gotas



Nucleolar Associating Domains (NADs)